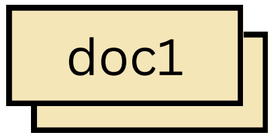
Natural Language Processing
Lab 7: Topic Models continued

Lonny Chen: l.chen@hertie-school.org
GRAD-E1282, Fall 2026
7-8 October 2026

BERTopic



Embed documents [sentence-transformers]

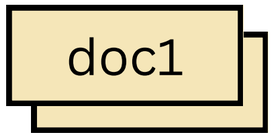
Reduce dimensions [UMAP]

Cluster docs into topics [HDBSCAN]

Vectorize clustered docs [CountVect.]

Weigh Terms by cluster [cTF-IDF]

BERTopic



Embed documents [sentence-transformers]

Reduce dimensions [UMAP]

Cluster docs into topics [HDBSCAN]

Vectorize clustered docs [CountVect.]

Weigh Terms by cluster [cTF-IDF]

384 dimensions

doc embedding

pre-trained sequence-aware embeddings

Documents

BERTopic

doc1

Embed documents [sentence-transformers]

Reduce dimensions [UMAP]

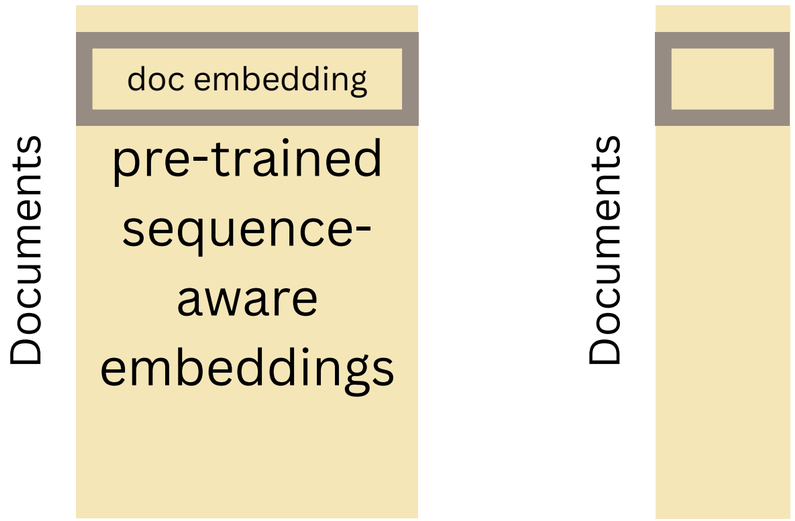
Cluster docs into topics [HDBSCAN]

Vectorize clustered docs [CountVect.]

Weigh Terms by cluster [cTF-IDF]

384 dimensions

5 dimensions



BERTopic

doc1

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384 dimensions

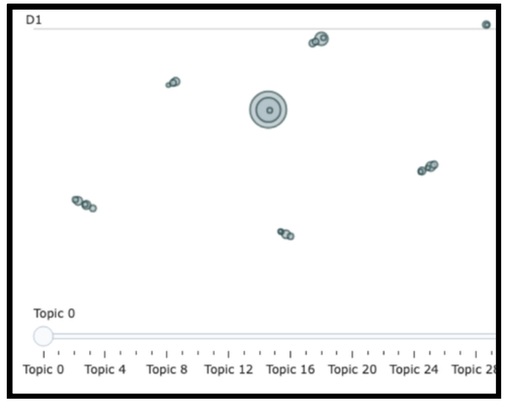
Documents

doc embedding

pre-trained sequence-aware embeddings

5 dimensions

Documents



ignore outliers (topic -1)

BERTopic

doc1

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384 dimensions

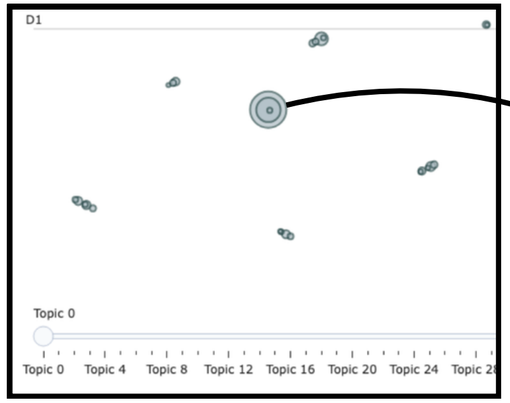
Documents

doc embedding

pre-trained sequence-aware embeddings

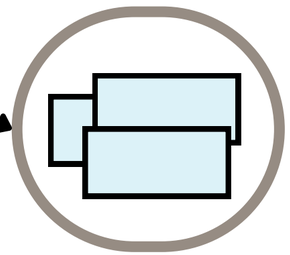
5 dimensions

Documents



ignore outliers (topic -1)

Topic n



BERTopic

doc1

Embed documents [sentence-transformers]

Reduce dimensions [UMAP]

Cluster docs into topics [HDBSCAN]

doc1

Vectorize clustered docs [CountVect.]

Weigh Terms by cluster [cTF-IDF]

384 dimensions

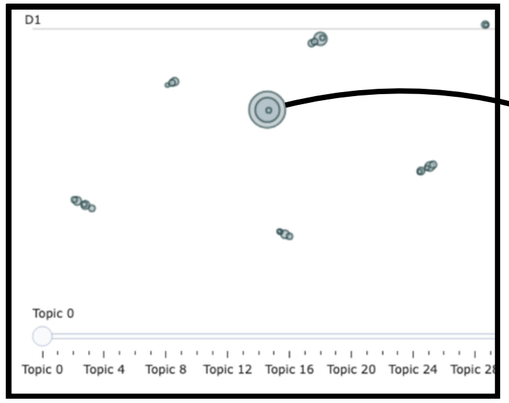
Documents

doc embedding

pre-trained sequence-aware embeddings

5 dimensions

Documents



ignore outliers (topic -1)

Topic n

Topics

Vocab terms

class TF-IDF weights

Coherence

```
'The heatwave pushed hospital admissions up as energy prices hit
records and drought hurt the harvest.'
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Coherence

```
'The heatwave pushed hospital admissions up as energy prices hit
records and drought hurt the harvest.'
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{NPMI}(w_i, w_j) = \frac{\ln\left(\frac{P(w_i,w_j)}{P(w_i)P(w_j)}\right)}{-\ln P(w_i, w_j)}$$

Coherence measures word similarity WITHIN a topic compared to a reference corpus.

Coherence

['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’, ‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Coherence

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave) *Pr(hospital)* *Pr(admissions)*

Pr(heatwave, hospital) *Pr(heatwave, admissions)* *Pr(hospital, admissions)*

Coherence

window_size = 5

```
['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices',
'hit', 'records', 'drought', 'hurt, 'harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave)	*Pr(hospital)*	*Pr(admissions)*
1 / 1	**1 / 1**	**1 / 1**

Pr(heatwave, hospital)	*Pr(heatwave, admissions)*	*Pr(hospital, admissions)*
1 / 1	**1 / 1**	**1 / 1**

Coherence

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave)

1 / 2

Pr(hospital)

2 / 2

Pr(admissions)

2 / 2

Pr(heatwave, hospital)

1 / 2

Pr(heatwave, admissions)

1 / 2

Pr(hospital, admissions)

2 / 2

Coherence

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave)	*Pr(hospital)*	*Pr(admissions)*
1 / 3	**3 / 3**	**3 / 3**

Pr(heatwave, hospital)	*Pr(heatwave, admissions)*	*Pr(hospital, admissions)*
1 / 3	**1 / 3**	**3 / 3**

Coherence

window_size = 5

```
['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices',
'hit', 'records', 'drought', 'hurt', 'harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave)	*Pr(hospital)*	*Pr(admissions)*
1 / 4	**3 / 4**	**4 / 4**

Pr(heatwave, hospital)	*Pr(heatwave, admissions)*	*Pr(hospital, admissions)*
1 / 4	**1 / 4**	**3 / 4**

Coherence

window_size = 5

```
['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices',
'hit', 'records', 'drought', 'hurt, 'harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave)	*Pr(hospital)*	*Pr(admissions)*
1 / 5	**3 / 5**	**4 / 5**

Pr(heatwave, hospital)	*Pr(heatwave, admissions)*	*Pr(hospital, admissions)*
1 / 5	**1 / 5**	**3 / 5**

Coherence

window_size = 5

```
['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices',
'hit', 'records', 'drought', 'hurt', 'harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave)	*Pr(hospital)*	*Pr(admissions)*
1 / 6	**3 / 6**	**4 / 6**

Pr(heatwave, hospital)	*Pr(heatwave, admissions)*	*Pr(hospital, admissions)*
1 / 6	**1 / 6**	**3 / 6**

Coherence

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest'
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Pr(heatwave)	*Pr(hospital)*	*Pr(admissions)*
1 / 7	**3 / 7**	**4 / 7**

Pr(heatwave, hospital)	*Pr(heatwave, admissions)*	*Pr(hospital, admissions)*
1 / 7	**1 / 7**	**3 / 7**

Coherence

window_size = 5

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{PMI}(w_i, w_j) = \ln\left(\frac{P(w_i, w_j)}{P(w_i)P(w_j)}\right)$$

PMI(heatwave, hospital) = ln ((1 / 7) / (1/7)(3/7)) = 0.8473

PMI(heatwave, admissions) = ln ((1 / 7) / (1/7)(4/7)) = 0.5596

PMI(hospital, admissions) = ln ((3 / 7) / (3/7)(4/7)) = 0.5596

Coherence

window_size = 5

['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices', 'hit', 'records', 'drought', 'hurt, 'harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{NPMI}(w_i, w_j) = \frac{\ln\left(\frac{P(w_i,w_j)}{P(w_i)P(w_j)}\right)}{-\ln P(w_i, w_j)}$$

NPMI(heatwave, hospital) = ln ((1 / 7) / (1/7)(3/7)) / -ln(1 / 7) = 0.4351

NPMI(heatwave, admissions) = ln ((1 / 7) / (1/7)(4/7)) / -ln(1 / 7) = 0.2871

NPMI(hospital, admissions) = ln ((3 / 7) / (3/7)(4/7)) / -ln(3 / 7) = 0.6216

Coherence

window_size = 5

['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices', 'hit', 'records', 'drought', 'hurt, 'harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{NPMI}(w_i, w_j) = \frac{\ln\left(\frac{P(w_i,w_j)}{P(w_i)P(w_j)}\right)}{-\ln P(w_i, w_j)}$$

NPMI(topic_0) = AVERAGE(NPMI(w_i, w_j)) = 0.4479

Coherence

window_size = 5

['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices', 'hit', 'records', 'drought', 'hurt, 'harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{NPMI}(w_i, w_j) = \frac{\ln\left(\frac{P(w_i,w_j)}{P(w_i)P(w_j)}\right)}{-\ln P(w_i, w_j)}$$

NPMI(topic_0) = AVERAGE(NPMI(w_i, w_j)) = 0.4479
NPMI(topic_1) = AVERAGE(NPMI(w_i, w_j)) = -0.4409

Coherence

window_size = 5

['heatwave', 'pushed', 'hospital', 'admissions', 'energy', 'prices', 'hit', 'records', 'drought', 'hurt, 'harvest']

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

$$\mathrm{NPMI}(w_i, w_j) = \frac{\ln\left(\frac{P(w_i,w_j)}{P(w_i)P(w_j)}\right)}{-\ln P(w_i, w_j)}$$

Coherence_NPMI = AVERAGE(NPMI(topic_k)) ~ 0.01

Diversity

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Diversity measures word uniqueness COMPARED TO other topics

Diversity

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’,
‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Diversity measures word uniqueness COMPARED TO other topics

Diversity = Unique words / Total words

Diversity

```
['heatwave’, ‘pushed’, ‘hospital’, ‘admissions’, ‘energy’, ‘prices’, ‘hit’, ‘records’, ‘drought’, ‘hurt, ’harvest']
```

```
topic_0 = ['heatwave', 'hospital', 'admissions'] # climate
topic_1 = ['heatwave', 'drought', 'harvest'], # health
```

Diversity measures word uniqueness COMPARED TO other topics

Diversity = 5 / 6 = 0.8333

Mid-semester check-in



Next Week: Neural Networks!

